

Software Engineering Project

Summary

This report was my coursework for a Software Engineering module at Heriot-Watt University (2nd year). This work was given 95%, one of the highest possible marks. It also accounted for 70% of my final mark in this module.

The objective of this individual project was to take an informal business remit for an automotive fleet maintenance system and take it through a rigorous software engineering design lifecycle. The system is designed to assist workshop mechanics with administrative tasks during vehicle servicing using a newly made workstation.

Below is the informal remit every student was given and everything beyond that is my own work, which is based on the remit.

Remit

SCA shall assist mechanics with the admin aspects of vehicle maintenance tasks. A mechanic will interface with SCA through a handheld QR-code reader, as well as a keyboard and screen, all of which are connected to a workstation. The workstation will be located next the tool-station within the maintenance workshop. In order to use SCA, an employee will have to be registered on the system. All registration must be performed by a supervisor. To complete the registration process, a supervisor is required to provide the following 4 data elements for the employee being registered: i) full name, ii) username, iii) work location, iv) status. Note that an employee's full name and username are to be represented as strings. A work location is represented by a five character alphanumeric code, e.g., AKT04, DUB01, EDI03, KUL01. In terms of status, there are four categories of user. That is, supervisor and three kinds of mechanic, i.e., apprentice, junior and senior. SCA should make use of an external system called Human Resources System (HRS) in order to validate the 4 data elements highlighted above. SCA should also create a unique Profile for each user that records these data elements. Note that SCA is required to internally manage a collection of user Profiles, i.e., an external database is not to be used. For an apprentice, their Profile should also capture their mentor, who must be a senior mechanic. It should also record the current stage within their apprenticeship, i.e., year-1, year-2, year-3 or year-4. For a senior mechanic, their Profile should also link to the Profiles of the apprentices that they are currently mentoring. In addition, it will record the date when they became a senior mechanic. For a junior mechanic, their Profile should record the date when they completed their apprenticeship and where they did their apprenticeship. Finally, a supervisor's Profile should record how long they have been a supervisor. Note that only supervisors should be allowed to manage a user's registration information. As part of the registration process, SCA should generate a unique 4-digit PIN. SCA should then share the PIN, via an email, with the user that is being registered. The generation of a PIN within SCA must reuse software from a related system, i.e., your design must reuse the skeleton Authenticator class as shown in Figure 1. As well as generating PINs, the Authenticator class must be refined in order to provide a user authentication' check. That is, for a given username and PIN, it should check that the pairing corresponds to a registered user of SCA. Your refined Authenticator class must be integrated

within your design for SCA. The process of servicing a vehicle starts with the mechanic scanning a QR-code that is located on the windscreen, i.e., the QR-code is printed on a sticky label that is provided by the company's fleet management team. The QR-code will provide SCA with a vehicle's, i) registration, ii) make, iii) model, iv) colour, and v) expected mileage. SCA must create a 'job record' for each vehicle once the QR-code is read. This job record must reuse a Job class that is used by a related system. The skeleton Job class that must be reused is shown in Figure 2. After a successful scan, SCA will prompt the mechanic with the 5 data items read from the QR-code. The mechanic is required to check these 5 data items when prompted by means of visual checks and confirm whether or not they are consistent. If a consistency check fails, then SCA should prompt the mechanic to provide additional information, i.e., a text-based explanation (500 characters including spaces max). Once a consistency check fails then the process of servicing the vehicle should be terminated. If all the consistency checks succeed, then the service proceeds with SCA prompting the mechanic to undertake a sequence of maintenance checks: firstly transmission-checks; secondly braking-checks; and lastly steering-checks. After each maintenance check the mechanic should be prompted to provide feedback via the workstation keyboard. Feedback on a check is either, i) passed-with-no-work-required, ii) passed-after-work-undertaken, iii) failed. Both ii) and iii) require the mechanic to provide additional information, i.e., text-based explanation (500 characters including spaces max). Note that only one user can be logged into a SCA workstation at a time. If a mechanic is logged out during the process of servicing a vehicle then the partial job record of the service will be lost and the process of servicing the vehicle would need to be started over, i.e., starting with the mechanic scanning the QR-code for the vehicle again. Note that only mechanics are allowed to perform the servicing process. A supervisor, however, can also be registered on SCA as senior mechanic, i.e., they can be registered separately as a senior mechanic. During the maintenance day (08:00–16:00), SCA is required to maintain internally a collection of job records (i.e., objects of the Job class) for each mechanic that has maintained vehicle(s) during the day. That is, the collection is associated with the mechanic, and only that mechanic. Daily, at 22:00, SCA is required to perform an automatic shutdown routine. Firstly, it should archive the day's job records, which requires SCA sending the job records to an external Job Archive System (JAS). Once JAS has acknowledged receiving the job records, SCA should delete its own copy. SCA should then store its current collection of user Profiles in a machine readable format on the workstation's solid state drive (SSD). Finally, SCA should store data related to user authentication in a machine readable format on the workstation's SSD before shunting itself down. Note that the startup routine for SCA should restore the user Profiles and user authentication data.

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D1 - Gap-Mitigation Table

ID	Gap	Mitigation
GM1	The system remit does not consider what would happen if the QR code fails to scan.	The system prompts the user with an alternative option to continue the servicing process by using the vehicles VIN number which the user would manually type in. VIN number is included in car details. The system will check if the VIN given matches any vehicles with the company. If a match is found, the user can continue with the process as usual.
GM2	The system remit does not state the process of how a PIN code is generated when a new user is registered.	The system uses a software that finds a random number between 0000-9999 that is not used by another user.
GM3	The system remit does not state what users can do if they forget their PIN number.	Provide a PIN reset mechanism at log in. The user can request their PIN to be reset, which requires supervisor authorisation. If approved, the system asks for an email which it will send the new PIN to. The previous PIN is invalidated and returned to the available pool of PINs.
GM4	The system remit does not define the behaviour if the system ran out of 4-digit PIN numbers.	For the purposes of this coursework, it is assumed that the business is medium-sized and will not exceed the available range of 4-digit PINs. If the system is upgraded in the future, the PIN length can be increased to 5 digits to provide a larger range of unique PINs.
GM5	The system remit does not state how the system knows what email to send the PIN to.	The system asks the user for their personal email, which then the system generates then sends the PIN. This email is not stored anywhere.
GM6	The system remit does not state how and where job records are stored throughout the day.	At 08:00, the collection is empty. During 08:00 – 16:00 (maintenance day), job records are stored as an in-memory collection. This collection is then dealt with appropriately at shutdown time.
GM7	The system remit does not consider if a vehicle passes without work done but may need future work.	Mechanics are asked if any potential future work should be noted after each maintenance check, regardless of if work is done or it passes/fails.
GM8	The system remit does not define what would happen if an incorrect PIN were entered.	System states PIN or username given is incorrect. The user is given 5 attempts before the account is locked for 10 minutes. If the user is locked out twice during the same day, the account is locked for 24 hours and an administrator is notified.
GM9	The system remit does not specify if usernames should be unique and what would happen in the case of duplicate usernames.	Every username should be unique. If a new user attempts to use a username that's already taken, a message is prompted stating that. Usernames cannot

		be changed and must be a length of 6-16 characters (as validated by HRS).
GM10	The system remit does not specify whether PINs can be manually changed.	PINs cannot be manually changed by any type of user.
GM11	The system remit does not consider if the email with the PIN does not send.	After an email is sent with the PIN, an option is given to the user to resend the email. If an attempt to resend the email is done over 3 times, the user is prompted with an email unavailable error where the error is logged, and an admin is notified to send a manual email.
GM12	The system remit does not specify whether the same vehicle can be serviced multiple times per day.	The system allows the same vehicle to be serviced more than once in the same day.
GM13	The system remit does not specify if supervisors are able to edit their own information or other supervisor information.	Supervisors can edit user registration details (except usernames) of any mechanic (Apprentice, Junior and Senior) at their work location and themselves. Supervisors cannot change user details of other supervisors.
GM14	The system remit does not consider if a service is being performed during 16:00 (end of maintenance day).	If a service has already started before 16:00, the user may complete it. No new services can be initiated after 16:00.
GM15	The system remit does not state what 'machine readable format' the user Profiles should be stored in.	All user profiles and user authentication information is converted into JSON (Java Script Object Notation) format.
GM16	The system remit does not specify whether "only one user is allowed to be log in in at one time" applies to all work locations or per work location.	Only one user is allowed to be logged in explicitly at each workstation. Therefore, one user at a time per workstation.
GM17	The system remit does not state what would happen if another user attempts to log in while another user is already logged in.	If a user attempts to log in on a workstation that already has an active session, the system displays an error message. There is no effect if a user tries to log in while another user is logged in at a different workstation.
GM18	The system remit does not state whether or whether not the system times out is a user does not interact with the system after a period.	After 20 minutes, the user will be prompted with a message asking if they want to continue. If the user keeps going, the process continues as usual. If the user wants to end the session, the user is automatically logged out, and any partial job record is deleted. This is done to save power, avoid other users using other accounts and to properly deal with all job records.
GM19	The system remit does not state how the system deals with supervisors also being able to be registered as a senior mechanic.	The system allows a supervisor to hold multiple roles (supervisor and senior only). After a senior is promoted to supervisor through the submit profile edit

		feature, a senior role can also be added to their profile. This means when a supervisor which is registered to both roles logs in, they pick what role they want for this session. This maintains integrity and flexibility for supervisors.
GM20	The system remit does not consider what happens if the user attempts to log in after 16:00 (end of maintenance day)	An error message is prompted to the user.
GM21	The system remit does not consider what happens if the user is logged in before 16:00 (end of maintenance day) and attempts to start a service after the time.	As users are automatically logged out at 16:00 if a service has not been begun, this problem will not arise.
GM22	The system remit does not consider what happens if any user profiles/user authentication data fails to convert into machine readable format before shutdown.	The shutdown process is aborted, the error is logged, the administrator is notified and retry conversion of any data that previously failed to convert. If data successfully converts, the system will then automatically shut down as usual.
GM23	The system remit does not consider what happens if a user attempts to log in to an account with missing information due to conversion failure.	An error message is given to the user, the error is logged and the administrator is notified.
GM24	The system remit does not mention if users are able to change personal details after registering.	Supervisors can request a profile edit, which is logged and must be approved by HRS.
GM25	The system remit does not mention what the process of deleting a user profile is.	The supervisors can request a profile to be deleted and can add additional information (up to 500 characters) to why they want to delete their profile.
GM26	The system remit does not mention when the system starts up.	The startup time for the system is 06:00 am.
GM27	The system remit does not mention how an email is sent to the user.	The system sends the given email to the Email Handling System (EHS), which is in charge of transferring the information given into an email and sending an email to the user.
GM28	The system remit does not specify how a user's log in details is validated.	The log in attempt is logged into a system file, and the details are passed to HRS to be validated.
GM29	The system remit does not explain how the system gains the information of who the new apprentice's mentor is.	After the registration details of the apprentice is entered and the HRS confirms the details, the Supervisor is prompted to input the apprentice's mentor (username) and the current stage of apprenticeship.
GM30	The system remit does not mention if the system checks if the mentor given is a senior mechanic as they should be.	When the Supervisor inputs the apprentice's mentor and apprenticeship stage, HRS validates the mentor is valid.

GM31	The system remit does not explain how the system will know the current stage the apprentice is in.	When an apprentice is registered, the Supervisor states the current stage the apprentice is in. On top of this, when a year has been completed by the apprentice, Supervisors are able to change the apprentice's current stage through the submit profile edit feature.
GM32	The system remit does not state how a user changes status.	A supervisor can change the status of users at its work location through the submit profile edit feature.
GM33	The system remit does not state what qualifies each user to each status.	Every new user can start at any status. When a user starts at Junior, the apprenticeship completion date and location when completed is NULL. When a user starts at Senior, the start date will be the current date when registered.
GM34	The system remit does not specify how it knows when an apprentice its apprenticeship.	When an apprentice is at year-4 on its profile, the Supervisor can change the apprentice to Junior status. When this happens, the current date and work location is stored and used to fill 'completion date' and 'location when completed' fields respectively.
GM34	The system remit does not define how the system should deal with knowing where they did their apprenticeship if an apprentice changes work location during the apprenticeship.	The concept of where the apprentice did its apprenticeship will be states in a 'location when completed' field. This will be the last location the junior was at before completing its apprenticeship.
GM35	The system remit does not specify is apprentices can skip years.	When an apprentice is registered, they can start at any stage the Supervisor would like. On top of this, an apprentice's current stage can be change at any time to any stage. This allows apprentices to skip years if the company believes they are progressing well.
GM36	The system remit does not state whether a user can be demoted status.	All demotions are done through the submit edit profile feature. A supervisor can only be demoted to a Senior. A senior can only be demoted to a Junior. A Junior cannot be demoted at all.
GM37	The system remit does not specify how the 'currently mentoring' list for Seniors is populated.	The 'currently mentoring' list begins as NULL and whenever an apprentice has that senior as a mentor (through registration or profile edit), that seniors list is updated to include that Apprentice.
GM38	The system remit does not state the process of determining how long a user has been a supervisor and if that information can be edited.	This field is set to 0 days as default and is determined when user information is restored by subtracting that current date by the date they became supervisor, which is also visible through their profile.
GM39	The system remit does not specify what the consistency checks are.	The 5 consistency checks that need to be performed at each service are: Check

		brake fluid level, check tire conditions, check coolant level, check wiper fluid level and check engine oil level.
GM40	The system remit does not specify the process for a mechanic giving feedback on a maintenance check.	After each check, the mechanic is prompted to put the feedback into a category (passed-with-no-work-required, passed-after-work-undertaken, failed), which then the additional information needed for that category will be available to be answered by the mechanic.
GM41	The system remit does not describe how the vehicle details are stored, including the QR code.	Every vehicle that is owned by the company is stored in a vehicle database externally. Every vehicle stored in the database have their own respective QR code and VIN number fields. SCA uses the Vehicle Identifying System (VIS) to match the given QR code/ VIN number with one in the database.

D2 – Functional and non-functional requirements

D2.1 - Abbreviation Table

Abbreviation	Meaning
HRS	Human Resources System
JAS	Job Archive System
SSD	Solid State Drive
JSON	Java Script Object Notation
EHS	Email Handling System
VIS	Vehicle Identifying System

D2.2 - Functional Requirements

ID	Functional Requirement
FR1	The system shall only permit registrations done by users with supervisor status.
FR2	The system shall capture these specific user registration details when registering a user: Full Name, Username, Work Location and Status.
FR3	The system shall use HRS to validate user registration details.
FR4	The system shall capture a new user's mentor when an apprentice is being registered.
FR5	The system shall prompt the Supervisor to enter a mentor and current stage when an apprentice is being registered, which then HRS validates that the mentor given is a senior.
FR6	The system shall ask for an email to send the PIN to when a new user is registered.
FR7	The system shall generate a unique 4-digit PIN when a new user is registered and ensure its not already used.
FR8	The system shall send the given email and generated PIN to EHS, which sends the user an automatic email.
FR9	The system shall store user information in-memory as a collection.
FR10	The system shall not allow PIN codes to be manually changed.
FR11	The system shall initially make newly promoted or registered supervisors just the supervisor role, which then they additionally be assigned with the senior role through the submit profile edit feature.
FR12	The system shall ask a supervisor that is also registered to the senior mechanic role what role they would like to proceed with after log in.
FR13	The system shall allow users to request a PIN reset at log in, which requires supervisor approval. The old PIN is invalidated, and the new PIN is sent to given email.
FR14	The system shall pass log in details to HRS to be validated.
FR15	The system shall log a log in attempt into a system file.
FR16	The system shall lock a user account after 5 consecutive failed login attempts for 10 minutes.
FR17	The system shall lock an account for 24 hours and notify an administrator if an account is locked twice in a day.
FR18	The system shall prompt users with an error message if a user attempts to log in while another user is already logged in at the same workstation.

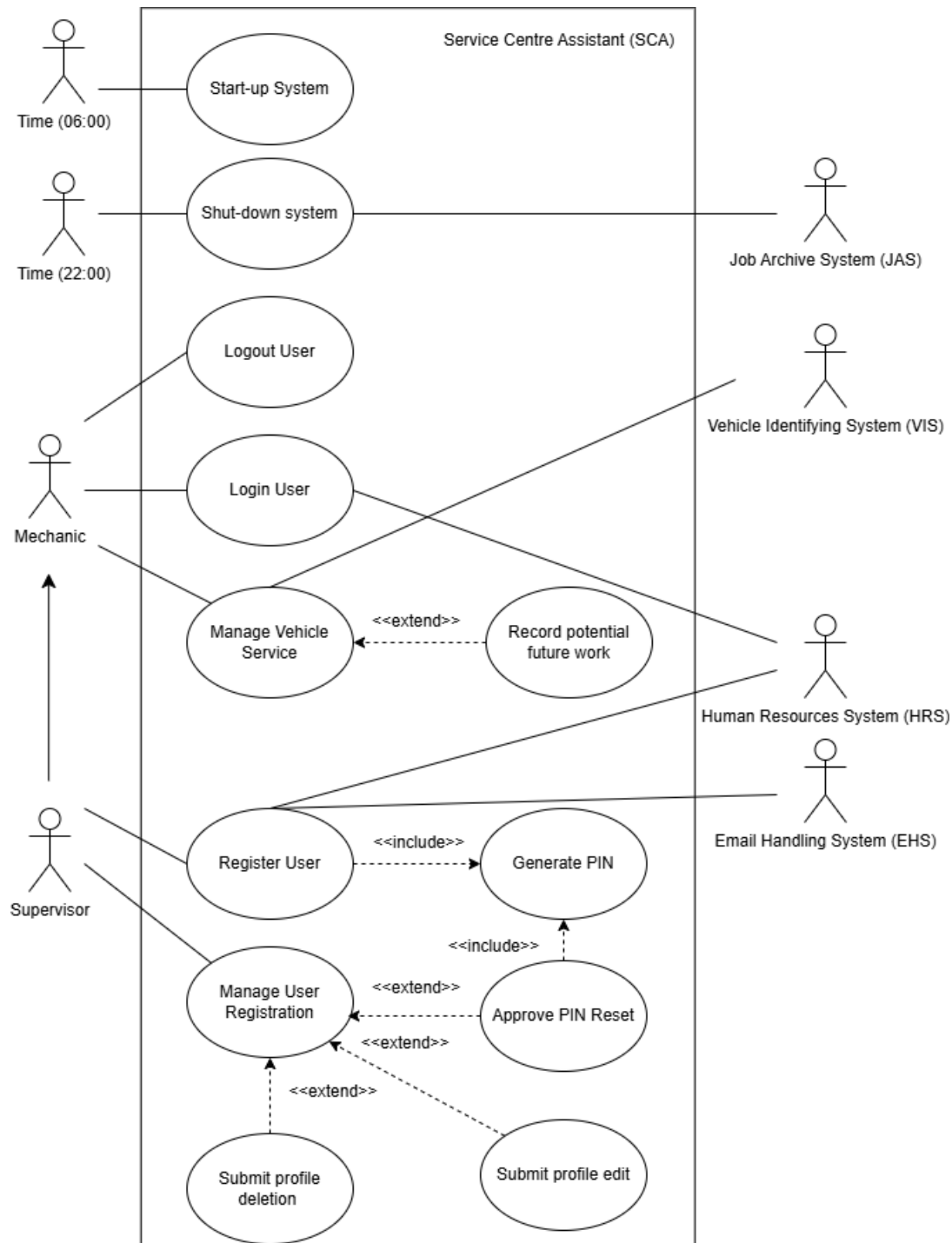
FR19	The system shall only give the user the option to service a car if they are logged in as an apprentice, junior or senior mechanic (not a supervisor).
FR20	The system shall prevent users from logging in after 16:00, prompting an error message if an attempt is made.
FR21	The system shall automatically log out all users at 16:00 (end of maintenance day) if no service has been initiated.
FR22	The system shall automatically log out all users immediately after a service is completed if the completion time is equal to or later than 16:00.
FR23	The system shall timeout the session after 20 minutes of inactivity. The user should be automatically logged out and any partial job record deleted.
FR24	The system shall detect if a QR code isn't being scanned properly and ensure a job record is not created.
FR25	The system shall validate the provided QR code or VIN number with VIS to ensure it matches with a company registered vehicle.
FR26	The system shall create a job record once the QR code has successfully been read.
FR27	The system shall prompt the user with the following 6 data items to visually check on the vehicle if a QR code read is successful: Registration, VIN, Make, Model, Colour and Expected Mileage.
FR28	The system shall ask the user to confirm if the data items are visually consistent with the information given.
FR29	The system shall ask the mechanic to perform the following 5 visual consistency checks after a job record is created: Check Brake Fluid, Check Wiper Fluid, Check Coolant Level, Check Engine Oil Level and Check condition of all tires.
FR30	The system shall prompt to give additional information if a consistency check fails.
FR31	The system shall prompt the user to give additional information up to 500 characters to describe why a vehicle failed if a visual consistency check fails.
FR32	The system shall terminate the process of servicing the car if a consistency check fails.
FR33	The system shall delete any partial job record and cancel servicing process if a consistency check fails.
FR34	The system shall enforce the following order of maintenance checks if all consistency checks pass: Transmission checks, Braking Checks and Steering Checks.
FR35	The system shall give the user these 3 options after each maintenance check: Passed with-no-work-required, Passed-after-work-undertaken or Failed.
FR36	The system shall prompt the user to give additional information if a maintenance check passes with work undertaken or fails.
FR37	The system shall give the user the option to add any potential future work after each check.
FR38	The system shall delete the partial job record if a user decides not to continue with the session or logs out.
FR39	The system shall allow the same vehicle to be serviced multiple times per day.
FR40	The system shall archive all job records collected throughout the day by sending the job records to JAS.
FR41	The system shall delete its copy of the job records once JAS acknowledge they have received a copy of the job records.
FR42	The system shall convert all existing user profiles into JSON format into one single file on the workstation's SSD after 22:00.
FR43	The system shall convert all user authentication data into JSON format into one single file on the workstation's SSD after 22:00.

FR44	The system shall abort shutdown, log the error, notify the administrator and retry conversion if any user profiles fail to convert to JSON format. If reconversion is successful, resume shutdown.
FR45	The system shall abort shutdown, log the error, notify the administrator and retry conversion if any user authentication data fails to convert to JSON format. If reconversion is successful, resume shutdown.
FR46	The system shall automatically shut down after it has deleted its copy of the day's job records and successfully converted and stored all user profiles and authentication data into their respective files.
FR47	The system shall restore all user profiles and user authentication data at 06:00 (start-up time).
FR48	The system shall log the error, notify the administrator and retry restoration if any user profiles fail to restore from JSON format to in-memory at startup.
FR49	The system shall log the error, notify the administrator and retry restoration if any user authentication data fails to convert from JSON to in-memory at startup.
FR50	The system shall prompt the user with an error message, log the error and notify the administrator if a user attempts to log into an account with any missing information due to conversion failure.
FR51	The system shall leave the 'date when completed' and 'location when completed' fields NULL if a user is demoted to or registered as a Junior.
FR52	The system shall use the current date for the 'senior start date' field when a user is registered or promoted to a senior.
FR53	The system shall leave the 'currently mentoring' field NULL when a user is registered, promoted or demoted to a Senior.
FR54	The system shall allow Supervisors to move apprentices to any current stage they would like through the submit profile edit feature (e.g year-1 to year-3 is allowed).
FR55	The system shall allow Supervisors to only be demoted to a Senior through the submit edit feature.
FR56	The system shall allow Seniors to only be demoted to a Junior through the submit edit feature.
FR57	The system shall ensure a Junior cannot be demoted to an Apprentice through the submit edit feature.
FR58	The system shall allow an Apprentice to be promoted to a Junior only if their stage is at year-4.
FR59	The system shall use the current date for the 'completion date' field when a user is promoted to a Junior.
FR60	The system shall add an apprentice to the seniors 'currently mentoring' list if that apprentice registers with or their profile is edited to include that senior as their mentor.
FR61	The system shall default 'supervisor for' field to 0 days and the current date for the 'supervisor since' field when a user is registered as or promoted to a supervisor.
FR62	The system shall subtract the current date from the supervisors 'supervisor since' date to determine the number of days the user has been a supervisor in the 'supervisor for' field.
FR63	The system shall allow Supervisors to submit a profile edit for any users at his work location, except other supervisors.
FR64	The system shall allow Supervisors to ask for the email to be resent.
FR65	The system shall prompt Supervisors with an email unavailable error, log the error and notify an admin to send a manual email when a fourth attempt to resend the email is made.

D2.3 - Non-Functional Requirements

ID	Non-Functional Requirement
NFR1	The system shall send generate and send a PIN to the given email within 5 seconds of the user completing the registration process.
NFR2	The system shall store, delete and update job records throughout the maintenance day without error.

D3 – Use Case Diagram



D4 – Use Case Specifications

Use Case: Register User
ID: 1
Goal: Successfully register a new user onto the system.
Primary Actor: Supervisor
Secondary Actor(s): Human Resources System (HRS), Email Handling System (EHS)
Preconditions: 1. The SCA is functioning correctly 2. There is a connection between SCA and HRS 3. Workstation logged into Supervisor role 4. Supervisor has user information available to put into system.
Postconditions: 1. New user account is in the system. 2. New user received confirmation and PIN code email.
Main Flow: 1. SCA prompts Supervisor to insert new user's details (full name, username, work location, work status and email). 2. Supervisor enters information. 3. SCA validates user information via HRS. 4. SCA creates new record for new user. 5. Include(GeneratePIN) 6. SCA sends newly generated PIN to EHS. 7. New user receives email from EHS with new PIN and registration details confirmation.
Alternative Flow: 1a... 7. Supervisor want to cancel user registration. 1. SCA asks supervisor to confirm cancellation or continue with registration process. 2. Continue to step 1a... 8.1a or 1a... 8.1b based on supervisor's input. 1a... 7.1a. Supervisor confirms cancellation. 1. SCA informs supervisor that user registration has been cancelled. 2. Use case terminates. 1a... 7.1b. Supervisor wants to continue. 1. Continue to step 1... 3 based on when case was initiated. 1b... 3. Supervisor is logged out during user registration before record is created. 1. SCA informs supervisor of logged out error and user registration is terminated. 2. Use case terminates. 2a. Session timeout during registration details entry. 1. SCA informs supervisor that user registration has timed out. 2. Use case terminates. 2b. Supervisor fails to enter mandatory registration details. 1. SCA informs supervisor that the empty field is missing information. 2. Return to step 2. 2b.2a, 3, 3a.2a. Supervisor enters invalid detail(s). 1. SCA informs supervisor invalid detail(s) were entered. 2. Supervisor re-enters information. 3. Return to step 2, 2b.2 or 3a.2 based on when case was initiated. 3a. Work status is apprentice. 1. SCA prompts Supervisor to insert apprentices mentor details (full name and username) and current stage (year-1 to year-4). 2. Supervisor enters information.

3. Return to step 3.
- 3b. Work status is junior.
 1. Set apprenticeship 'completion date' and 'location of completion' to NULL.
 2. Continue to step 4.
- 3c. Work status is senior.
 1. Set 'mentoring list' to NULL and 'senior start date' to current date.
 2. Continue to step 4.
- 3d. Work status is supervisor.
 1. Set 'supervisor for' to 0 days and 'supervisor since' to current date.
 2. Continue to step 4.
- 3e. HRS system unavailable for validation.
 1. SCA prompts supervisor with validation unavailable error.
 2. SCA logs error and admin is notified.
 3. Use case terminates.
- 3f. Supervisor gives duplicate detail(s).
 1. SCA informs supervisor that a specific field entered already exists.
 2. Return to step 2.
- 4a. System fails to create new record.
 1. SCA prompts supervisor with record creation error.
 2. SCA logs error and notifies admin.
 3. Use case terminates.
- 4b... 7. Supervisor is logged out during user registration after record is created.
 1. SCA informs supervisor of logged out error and that new user should receive confirmation and PIN code email and if not, PIN can be reset.
 2. Attempt to complete use case.
 3. Use case terminates.
- 4c... 7. Supervisor continues with inaccurate detail(s).
 1. Supervisor clicks return button.
 2. SCA deletes user record from collection.
 3. Return to step 2.
- 5a. SCA fails to generate new PIN.
 1. SCA prompts supervisor with PIN generation error.
 2. SCA deletes partial user record.
 3. SCA logs error and notifies admin.
 4. Use case terminates.
- 6a. SCA fails to send newly generated PIN to EHS.
 1. SCA prompts supervisor with email error, but that the new user has been successfully created and to wait for future emails with PIN and confirmation.
 2. SCA logs error and notifies admin.
 3. Use case terminates.
- 7a. Email fails to send to new user.
 1. SCA informs Supervisor of email error.
 2. SCA increments failed email attempts counter.
 3. Return to step 7.
- 7b. Email fails to send to new user (fourth attempt)
 1. SCA prompts Supervisor with email unavailable error.
 2. SCA prompts that user has been successfully created and to wait for future emails with PIN and confirmation.
 3. SCA logs error and notifies admin.
 4. Use case terminates.
- 7c. Email was sent but atleast one failed email attempt was made.
 1. Reset failed email attempt counter.

Use Case: Manage Vehicle Service

ID: 2

Goal: Successfully document a vehicle service onto a newly created job record.

Primary Actor: Mechanic

Secondary Actor(s): Vehicle Identifying System (VIS)

Preconditions:

1. SCA is functioning correctly.
2. Workstation is logged into mechanic role.
3. Vehicle is in same location as workstation and ready to be serviced.

Postconditions:

1. Vehicle Details are updated.
2. Vehicle is serviced.
3. Job record is created and populated with maintenance feedback.

Main Flow:

1. Mechanic successfully identifies vehicle.
2. SCA creates partial job record for vehicle and stores it locally.
3. SCA asks mechanic to confirm vehicle details.
4. Mechanic confirms vehicle details are correct.
5. SCA prompts mechanic to perform and record visual consistency checks (brake fluid level, coolant fluid level, engine oil level, wiper fluid level, condition of tires).
6. Mechanic confirms all consistency checks passed.
7. SCA informs mechanic to perform transmission-checks.
8. Mechanic provides feedback on check (feedback category).
9. SCA asks mechanic if any potential future work was identified.
10. Mechanic provides potential future work.
11. SCA informs mechanic to perform braking-checks.
12. Mechanic provides feedback on check (feedback category).
13. SCA asks mechanic if any potential future work was identified.
14. Mechanic provides potential future work.
15. SCA informs mechanic to perform steering-checks.
16. Mechanic provides feedback on check (feedback category).
17. SCA asks mechanic if any potential future work was identified.
18. Mechanic provides potential future work.
19. SCA moves finished job record into internal collection of job records.

Alternative Flow:

- 1a. Session timeout before vehicle has been identified.
 1. SCA informs mechanic that session has timed out.
 2. Use case terminates.
- 1b. Mechanic chooses to identify with QR code.
 1. Mechanic selects "Identify via QR code".
 2. SCA prompts mechanic to give a QR code to read using camera.
 3. Mechanic scans QR code.
 4. SCA validates QR code with VIS.
 5. SCA prompts mechanic of successful match.
- 1b.3a. SCA cannot read a QR code after 15 seconds.
 1. SCA prompts mechanic with scanning error and to try again.
 2. Return to step 1b.2.
- 1b.4a. QR code fails to send to VIS.
 1. SCA prompts mechanic with vehicle identification system error and suggests identifying the vehicle with its VIN number.
 2. SCA logs error and notify admin.
 3. Return to step 1.

- 1b.4b. QR code given fails to match with any registered vehicle.
 - 1. SCA prompts mechanic with incorrect QR code error and suggests to try again.
 - 2. Return to step 1.
- 1b.4c. Invalid QR code.
 - 1. SCA prompts mechanic with incorrect QR code error and suggests to try again.
 - 2. Return to step 1.
- 1c. Mechanic chooses to identify with VIN number.
 - 1. Mechanic selects "Identify via VIN number".
 - 2. SCA prompts mechanic to type vehicles VIN number.
 - 3. Mechanic types VIN number.
 - 4. SCA validates VIN number with VIS.
 - 5. SCA informs mechanic of successful match.
- 1c.4a. Invalid VIN Format.
 - 1. SCA prompts mechanic with invalid VIN error and to try again.
 - 2. Return to step 1.
- 1c.4b. VIN Not Found.
 - 1. SCA prompts mechanic with vehicle not found error and to try again.
 - 2. Return to step 1.
- 1c.4c. VIN Number fails to send to VIS.
 - 1. SCA prompts mechanic with vehicle identification system error and suggests to try again.
 - 2. SCA logs error and notifies admin.
 - 3. Return to step 1.
- 1d. Mechanic logged out before vehicle is identified.
 - 1. SCA informs mechanic of log out error.
 - 2. Use case terminates.
- 2a... 18. Mechanic wants to cancel servicing process.
 - 1. SCA asks mechanic to confirm cancellation or continue with registration process.
 - 2. Continue to step 2a... 18.1a. or 2a... 18.1b. based on mechanic's input.
- 2a... 18.1a. Mechanic confirms they want to cancel.
 - 1. SCA informs mechanic that servicing process has been cancelled.
 - 2. SCA deletes partial job record.
 - 3. Use case terminates.
- 2a... 18.1b. Mechanic wants to continue.
 - 1. Return to step 2a... 18 based on when case was initiated.
- 2b... 18. Session timeout after vehicle has been identified.
 - 1. SCA informs mechanic that session has timed out.
 - 2. SCA deletes partial job record.
 - 3. Use case terminates.
- 2c... 18. Mechanic is logged out during servicing process.
 - 1. SCA informs mechanic of logged out error and servicing process is terminated.
 - 2. SCA deletes partial job record.
 - 3. Use case terminates.
- 2d. SCA fails to create job record.
 - 1. SCA informs mechanic of record creation error.
 - 2. SCA logs error and notifies admin.
 - 3. Use case terminates.
- 4a. Any car detail is incorrect.
 - 1. Mechanic clicks "Edit car details".
 - 2. Mechanic clicks desired field to change.
 - 3. Mechanic enters new information.
 - 4. Mechanic confirms new details.
 - 5. Return to step 4.

- 4a.3a. Mechanic enters invalid new car detail(s).
 - 1. SCA informs mechanic that new detail(s) given is invalid.
 - 2. Return to step 4a.
- 6a. Any consistency check fails.
 - 1. SCA informs mechanic that due to consistency check fail, servicing process is terminated.
 - 2. SCA deletes partial job record.
 - 3. Use case terminates.
- 8a., 12a., 16a. Feedback on check is "passed-after-work-undertaken" or "failed".
 - 1. SCA prompts mechanic to provide information on what work was done on the vehicle.
 - 2. Mechanic provides information.
 - 3. Continue to step 9, 13 or 17 based on when case was initiated.
- 8a.2a., 10a., 12a.2a., 14a., 16a.2a., 18a. Mechanic enters information over 500 characters.
 - 1. SCA prompts mechanic of too many characters error and to try again.
 - 2. Return to step 8a, 10a, 12a., 14a, 16a, or 18a. based on when case was initiated.
- 8a.2a., 10b., 12a.2a., 14b., 16a.2b., 18b. Mechanic leaves information field blank.
 - 1. SCA prompts mechanic of empty field error and to try again.
 - 2. Return to step 8a, 10b, 12a, 14b, 16a, or 18b based on when case was initiated.
- 9a., 13a., 17a. Mechanic found no potential future work.
 - 1. Continue to step 10, 14 or 18 based on when case was initiated.
- 10c., 14c., 18c. Mechanic accidentally pressed yes on potential future work.
 - 1. Mechanic presses return button.
 - 2. Return to step 9, 13 or 17 based on when case was initiated.
- 19a. SCA fails to move final job record to collection.
 - 1. SCA informs mechanic of system memory error.
 - 2. SCA keeps final job record in local memory.
 - 3. SCA logs error and notifies admin.
 - 4. Return to step 19.
- 19a.a. Three attempts have already been made to relocate final job record.
 - 1. SCA informs mechanic that due to system error, services cannot currently be made.
 - 2. SCA disables servicing process on that specific workstation, keeps final job record in local memory for when problem is fixed.

D5 – Traceability Matrix

D5.1 - Abbreviation Table

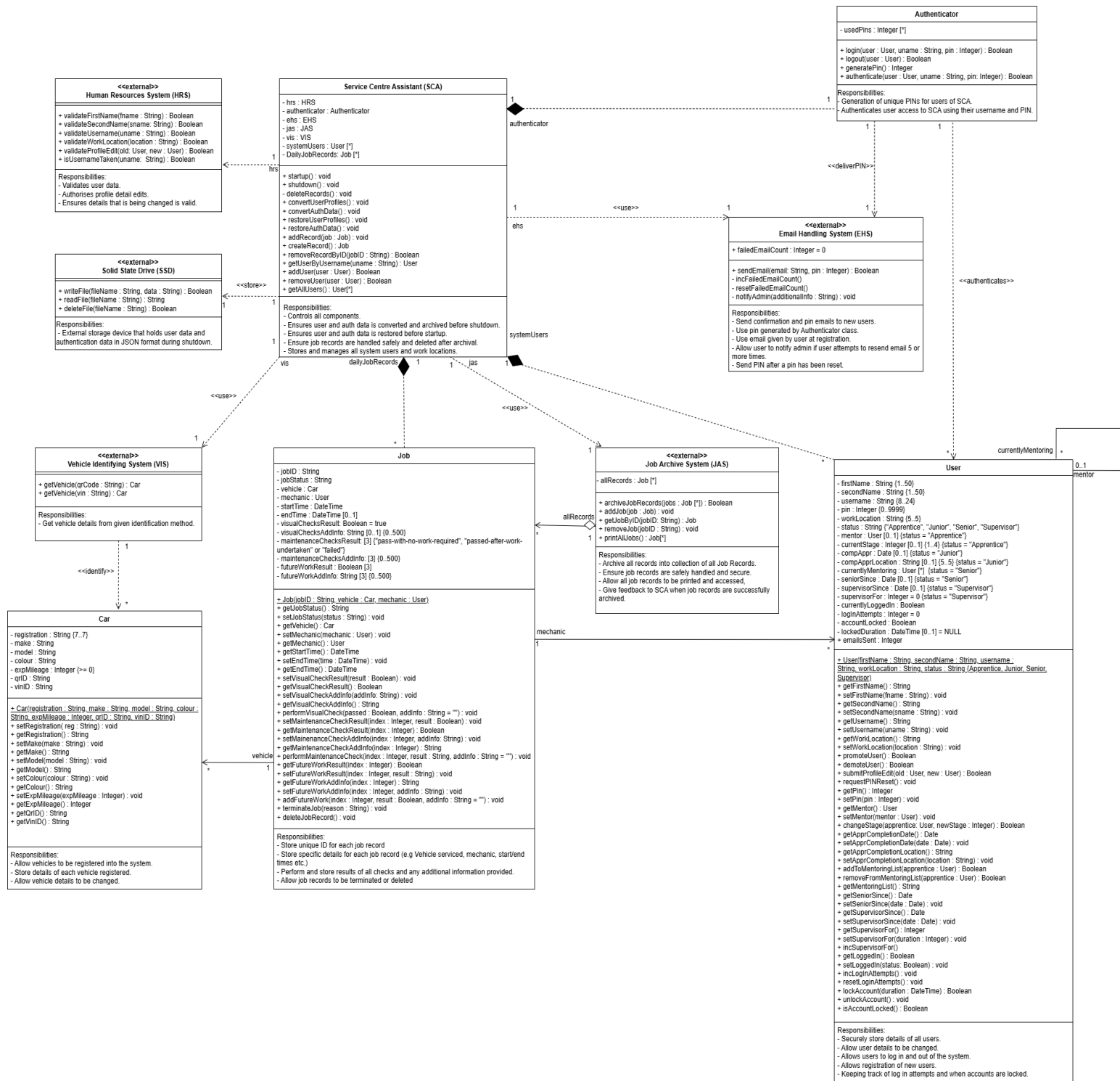
Abbreviation	Specific Use case
UC1	Start-up System
UC2	Shut-down System
UC3	Login User
UC4	Logout User
UC5	Register User
UC6	Manage User
UC7	Manage Vehicle Service

D5.2 - Traceability Matrix

FR ID	BASE USE CASE						
	UC1	UC2	UC3	UC4	UC5	UC6	UC7
FR1					X		
FR2					X		
FR3					X		
FR4					X		
FR5					X		
FR6					X		
FR7					X		
FR8					X		
FR9					X		
FR10					X		
FR11					X		
FR12			X				
FR13			X				
FR14			X				
FR15			X				
FR16			X				
FR17			X				
FR18			X				
FR19							X
FR20			X				
FR21				X			
FR22				X			
FR23				X			
FR24							X
FR25							X
FR26							X

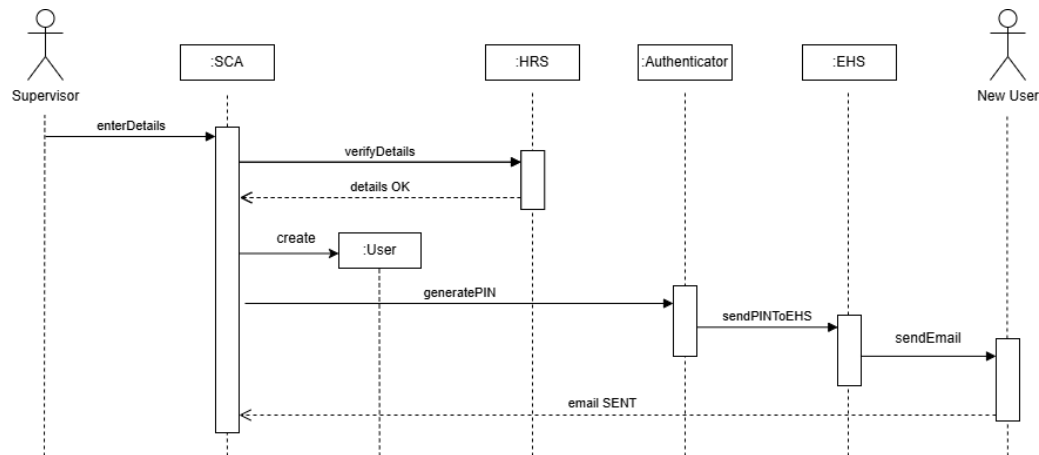
FR27							X
FR28							X
FR29							X
FR30							X
FR31							X
FR32							X
FR33							X
FR34							X
FR35							X
FR36							X
FR37							X
FR38							X
FR39							X
FR40		X					
FR41		X					
FR42		X					
FR43		X					
FR44		X					
FR45		X					
FR46		X					
FR47	X						
FR48	X						
FR49	X						
FR50	X						
FR51							
FR52					X		
FR53					X		
FR54					X		
FR55					X		
FR56					X		
FR57					X		
FR58					X		
FR59					X		
FR60					X		
FR61					X		
FR62					X		
FR63					X		
FR64					X		
FR65					X		

D6 – Class Diagram



D7 – Sequence Diagrams

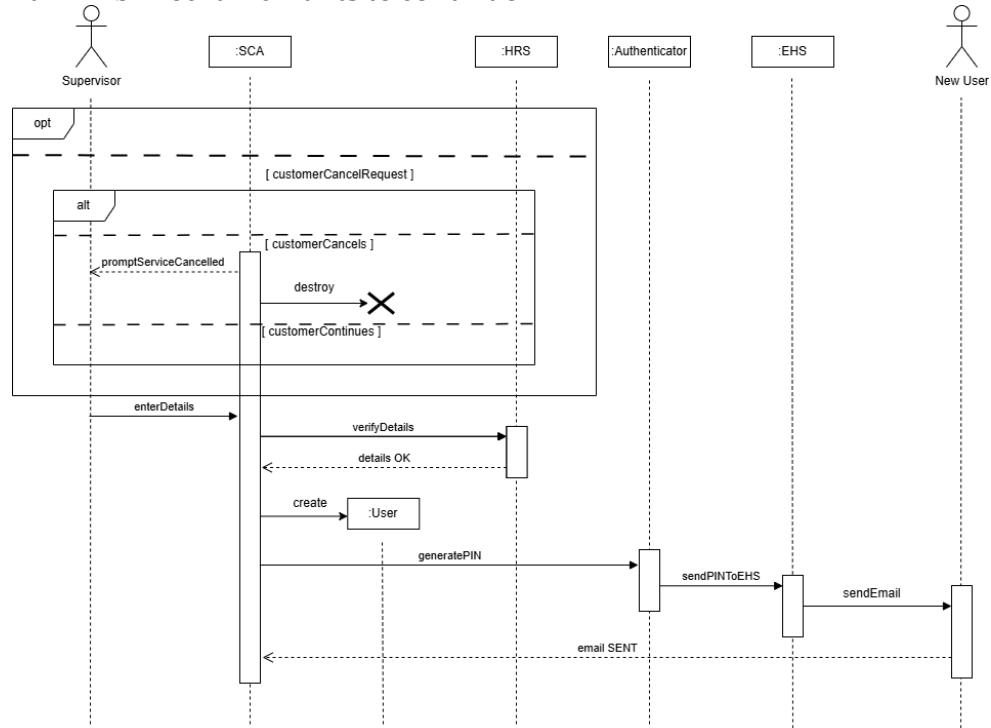
1.. 7 Main Flow



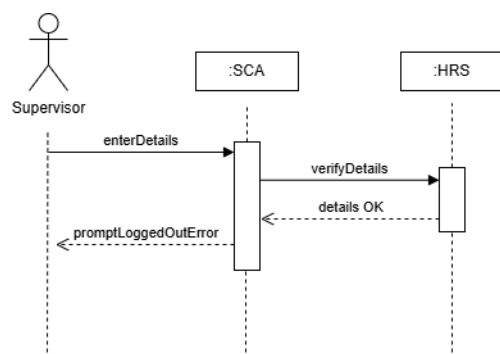
1a... 7. Mechanic wants to cancel user registration.

1a... 7.1a. Mechanic confirms they want to cancel.

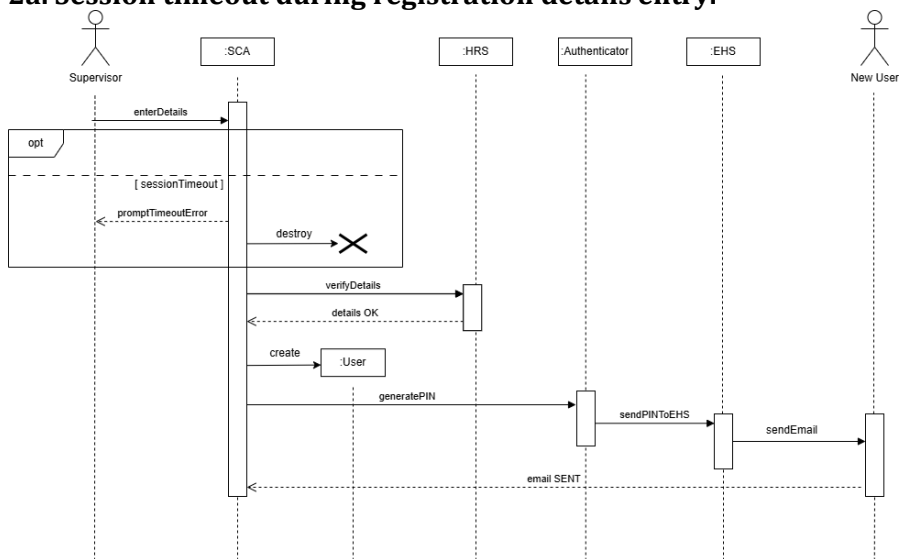
1a... 7.1b. Mechanic wants to continue.



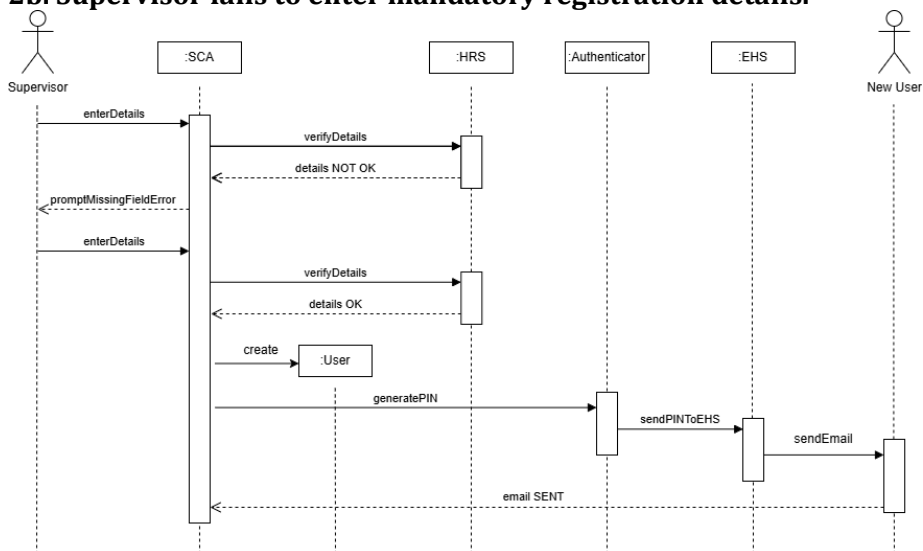
1b... 3. Supervisor is logged out during user registration before record is created.



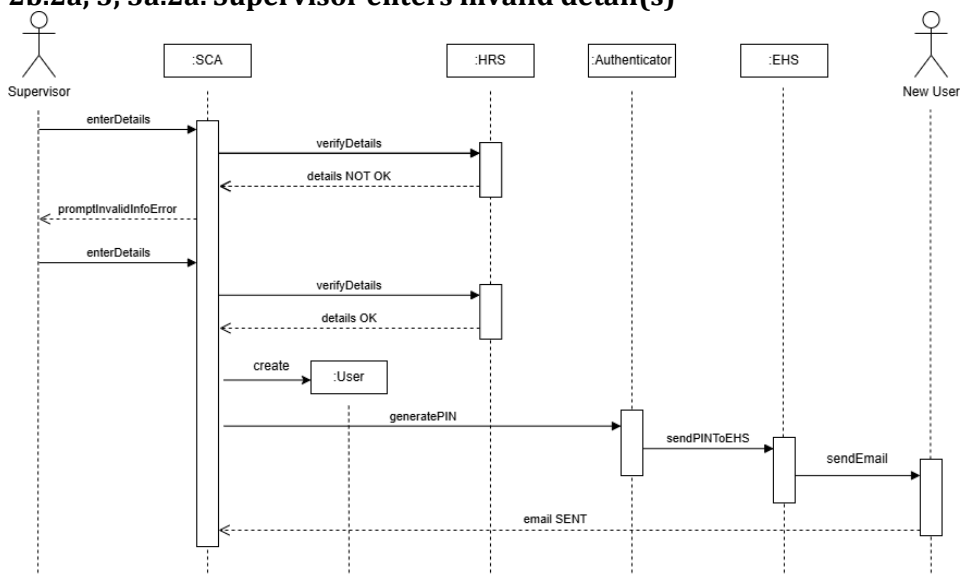
2a. Session timeout during registration details entry.



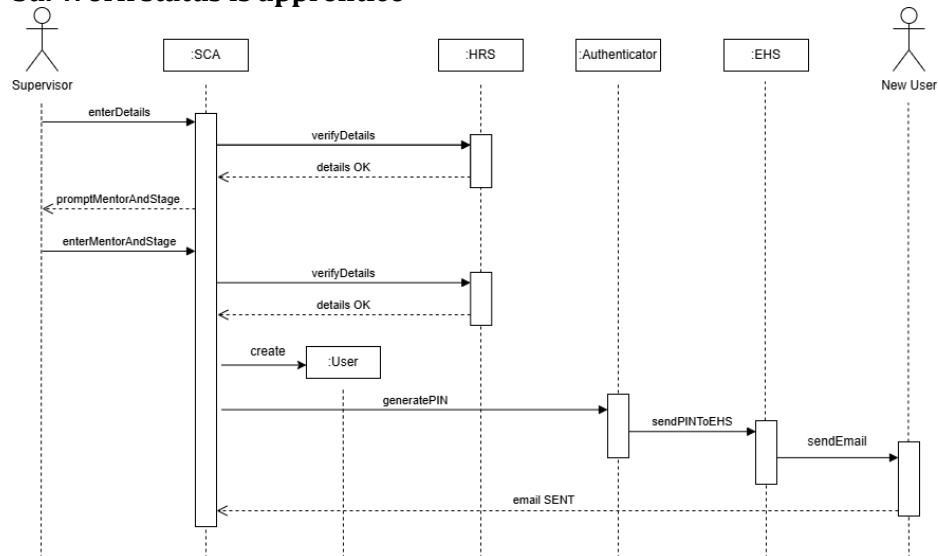
2b. Supervisor fails to enter mandatory registration details.



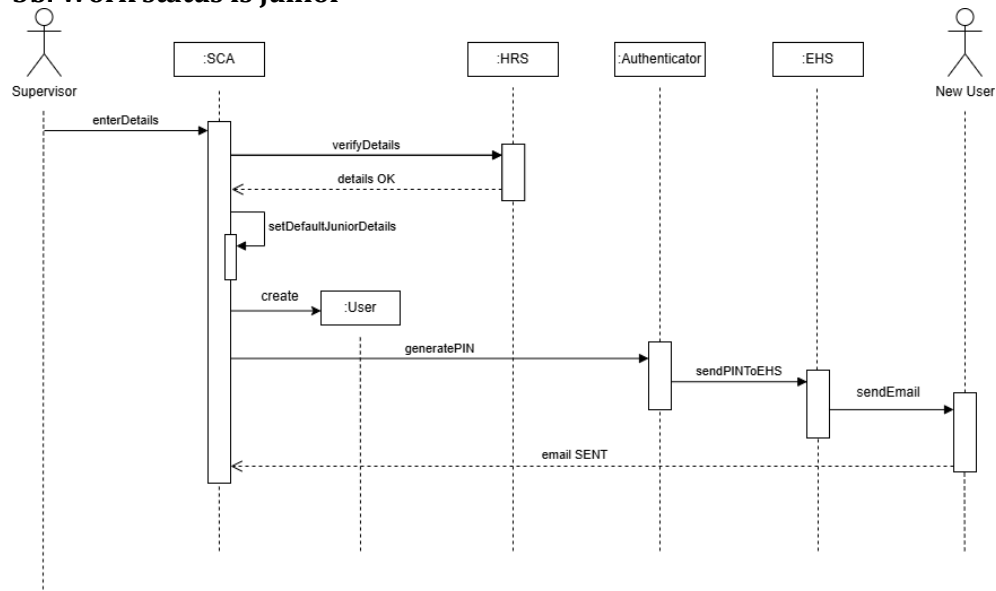
2b.2a, 3, 3a.2a. Supervisor enters invalid detail(s)



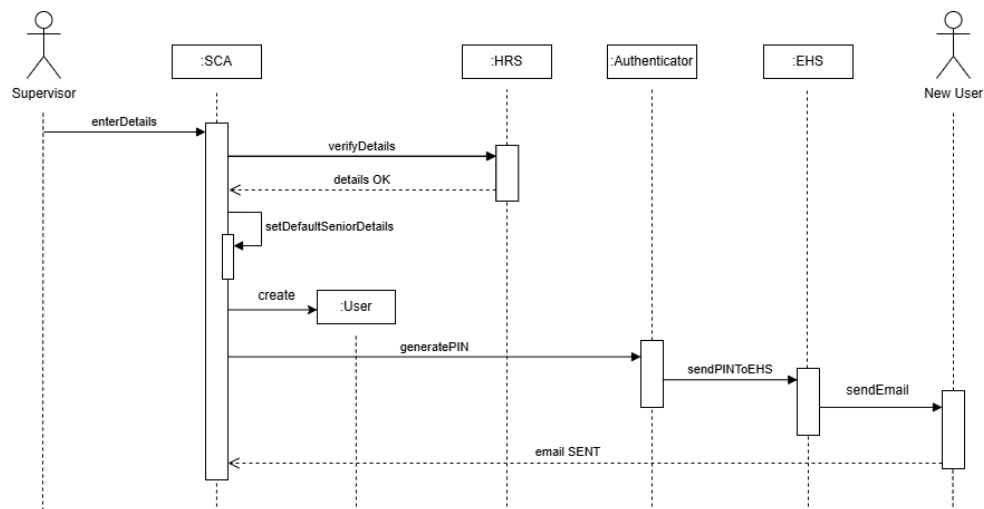
3a. Work status is apprentice



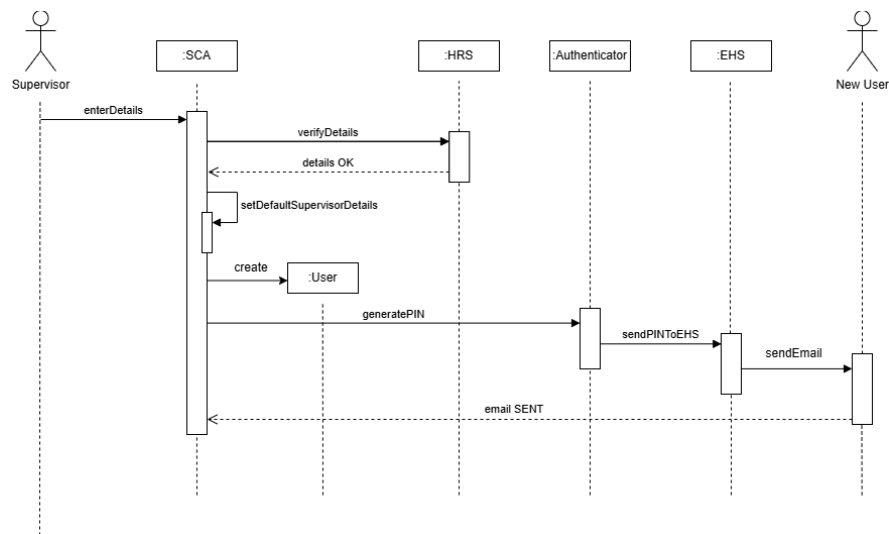
3b. Work status is junior



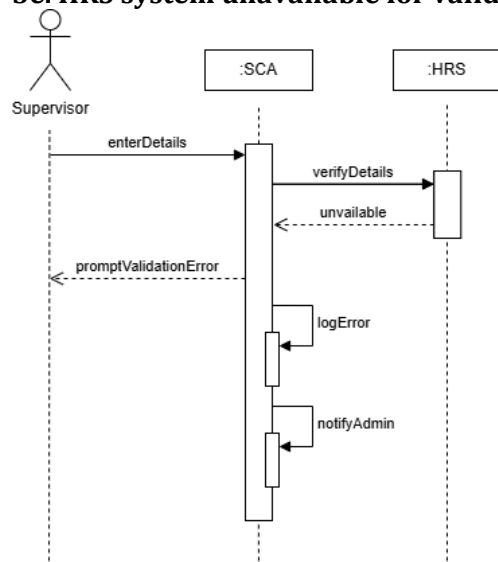
3c. Work status is senior



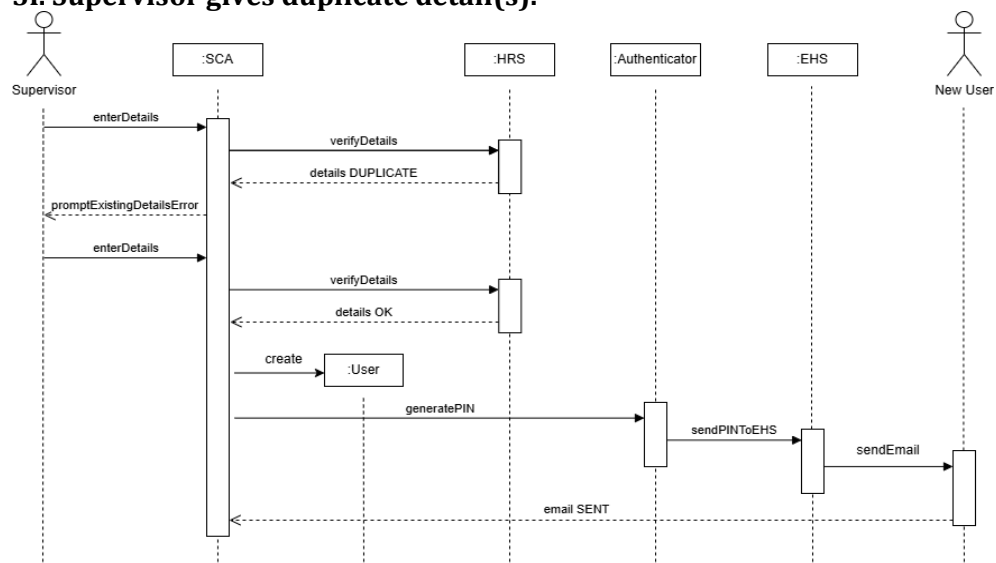
3d. Work status is supervisor.



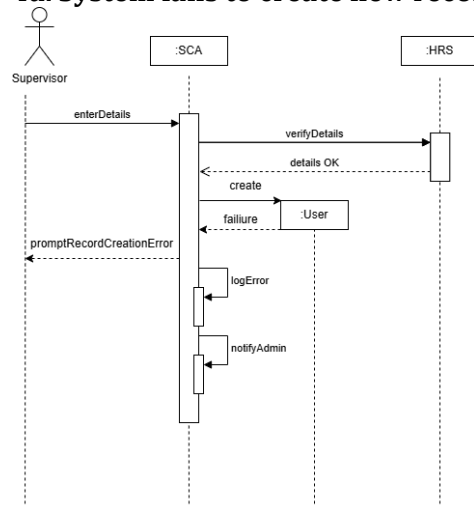
3e. HRS system unavailable for validation.



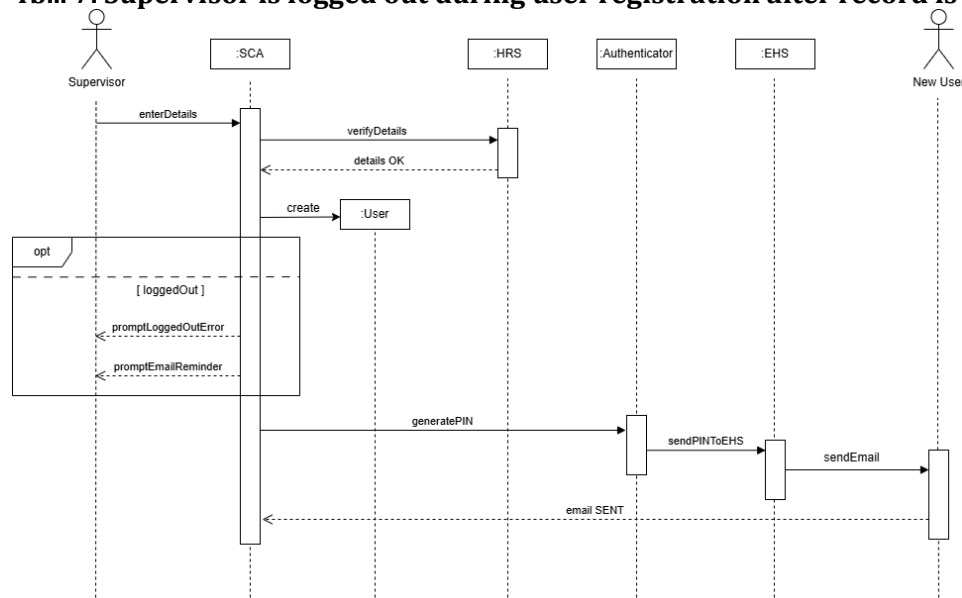
3f. Supervisor gives duplicate detail(s).



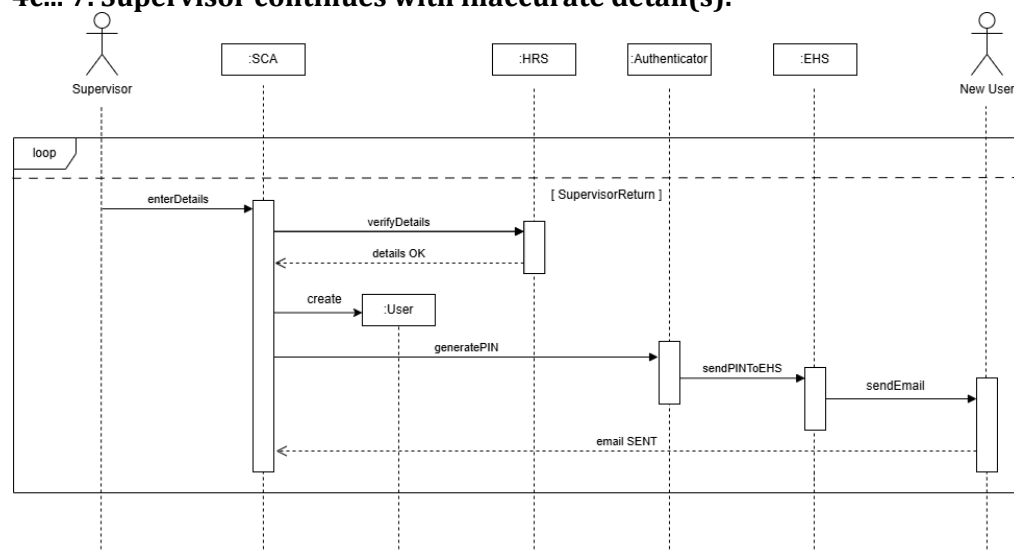
4a. System fails to create new record.



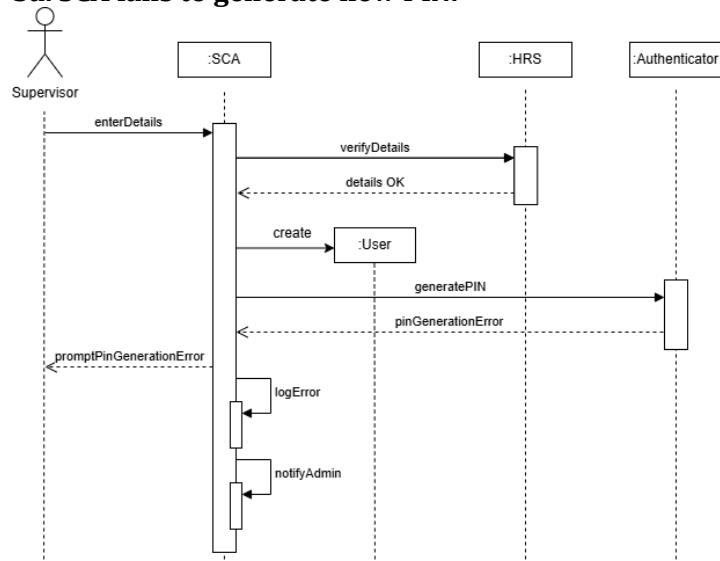
4b... 7. Supervisor is logged out during user registration after record is created.



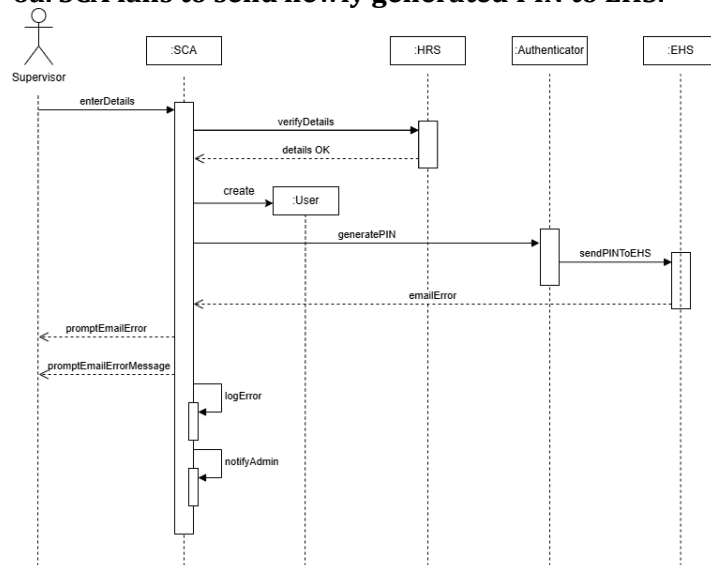
4c... 7. Supervisor continues with inaccurate detail(s).



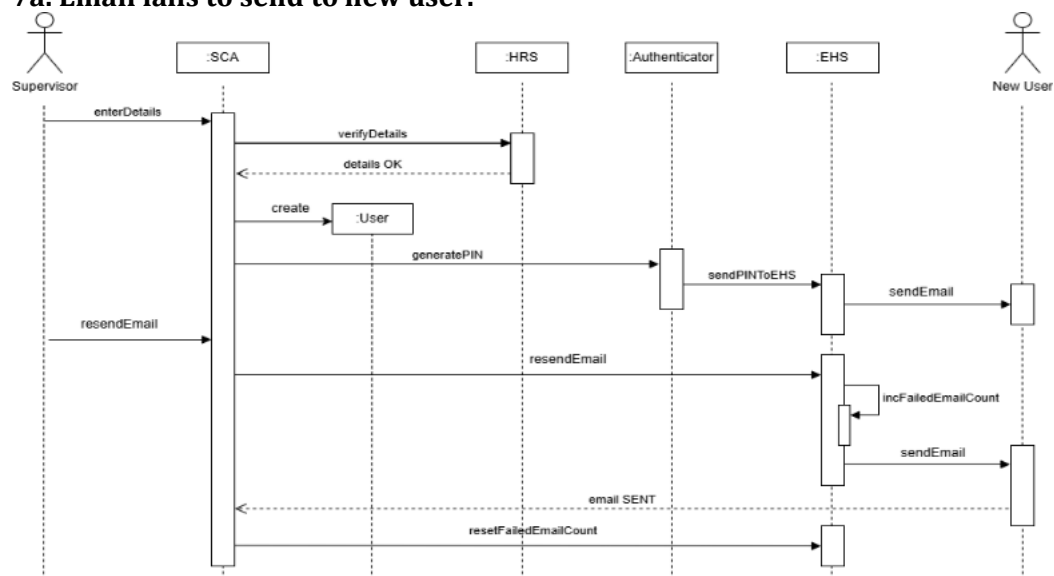
5a. SCA fails to generate new PIN.



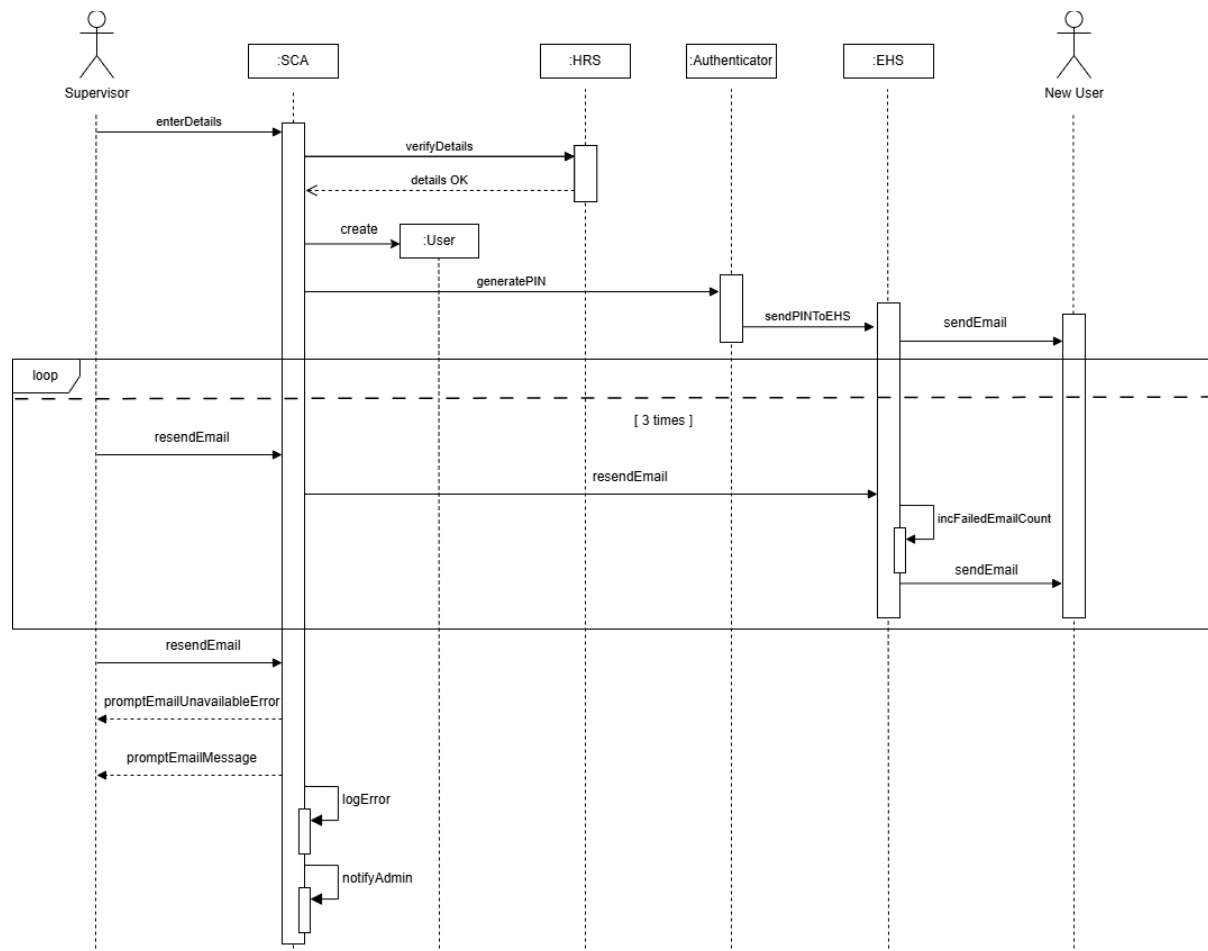
6a. SCA fails to send newly generated PIN to EHS.



7a. Email fails to send to new user.



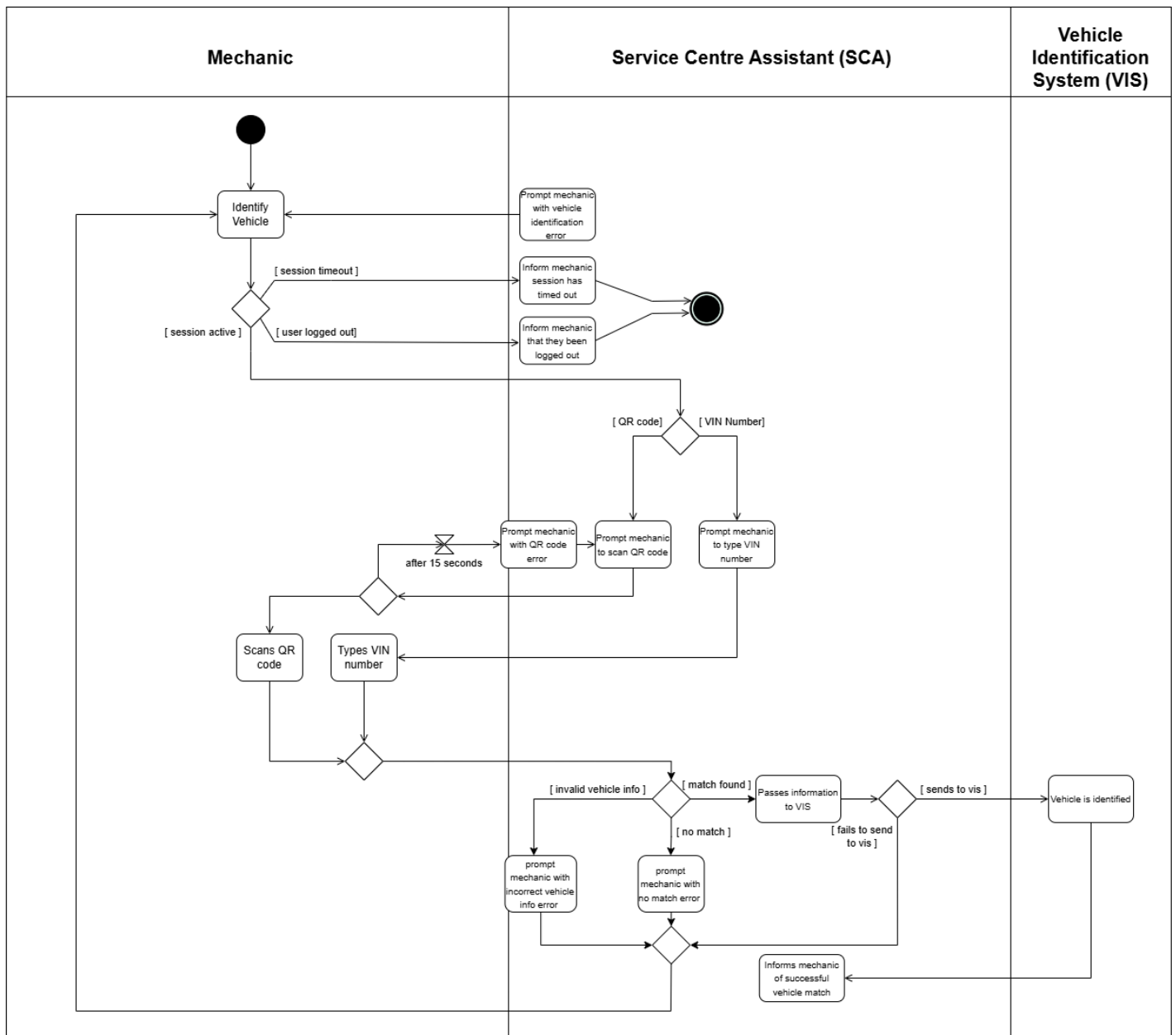
7b. Three failed attempts to send an email has been made.



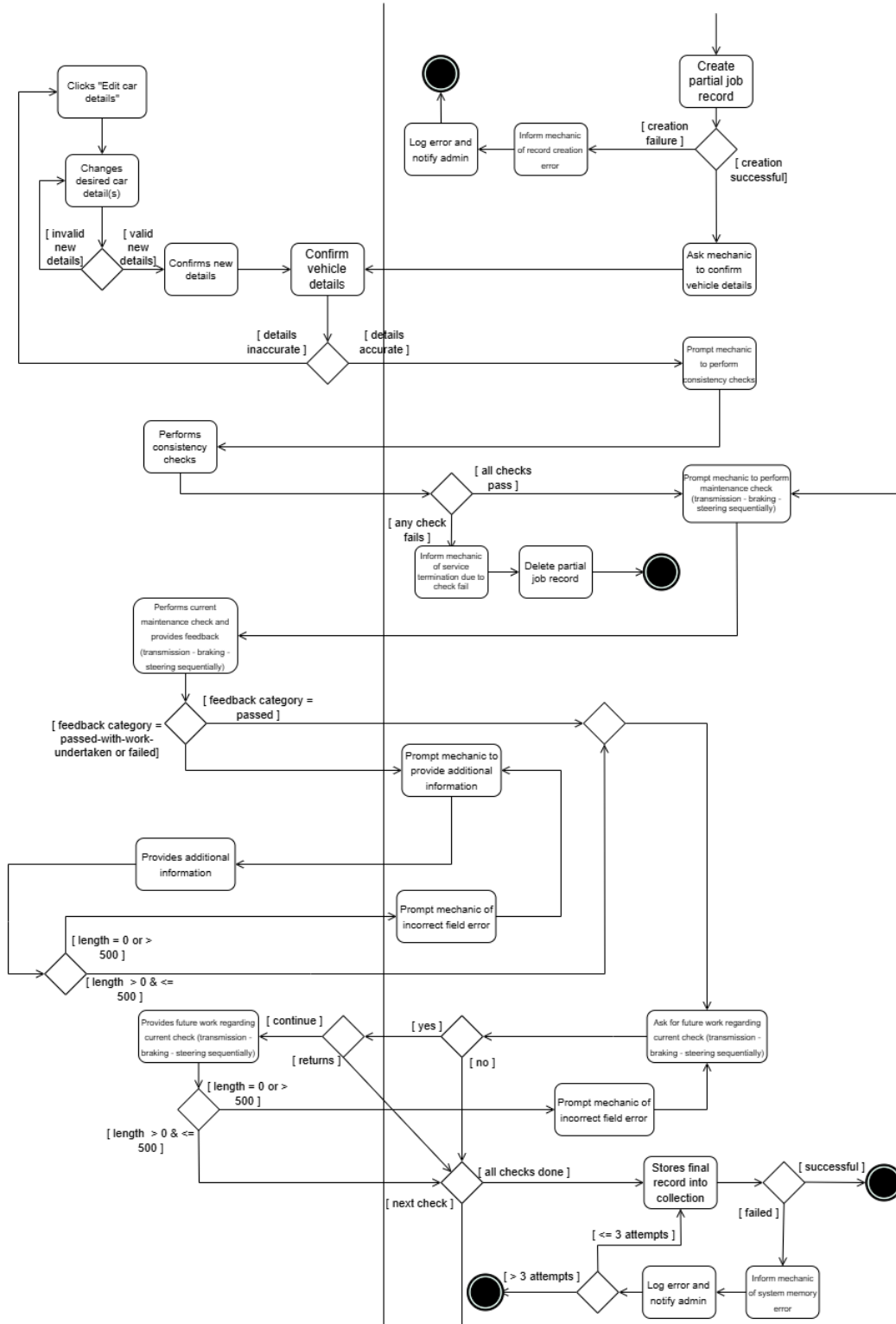
7c. Email was sent but at least one failed email attempt was made. - Treated as an auxiliary flow for 7a, so this flow is represented in sequence diagram 7a (resets failed email attempt counter).

D8 – Activity Diagram

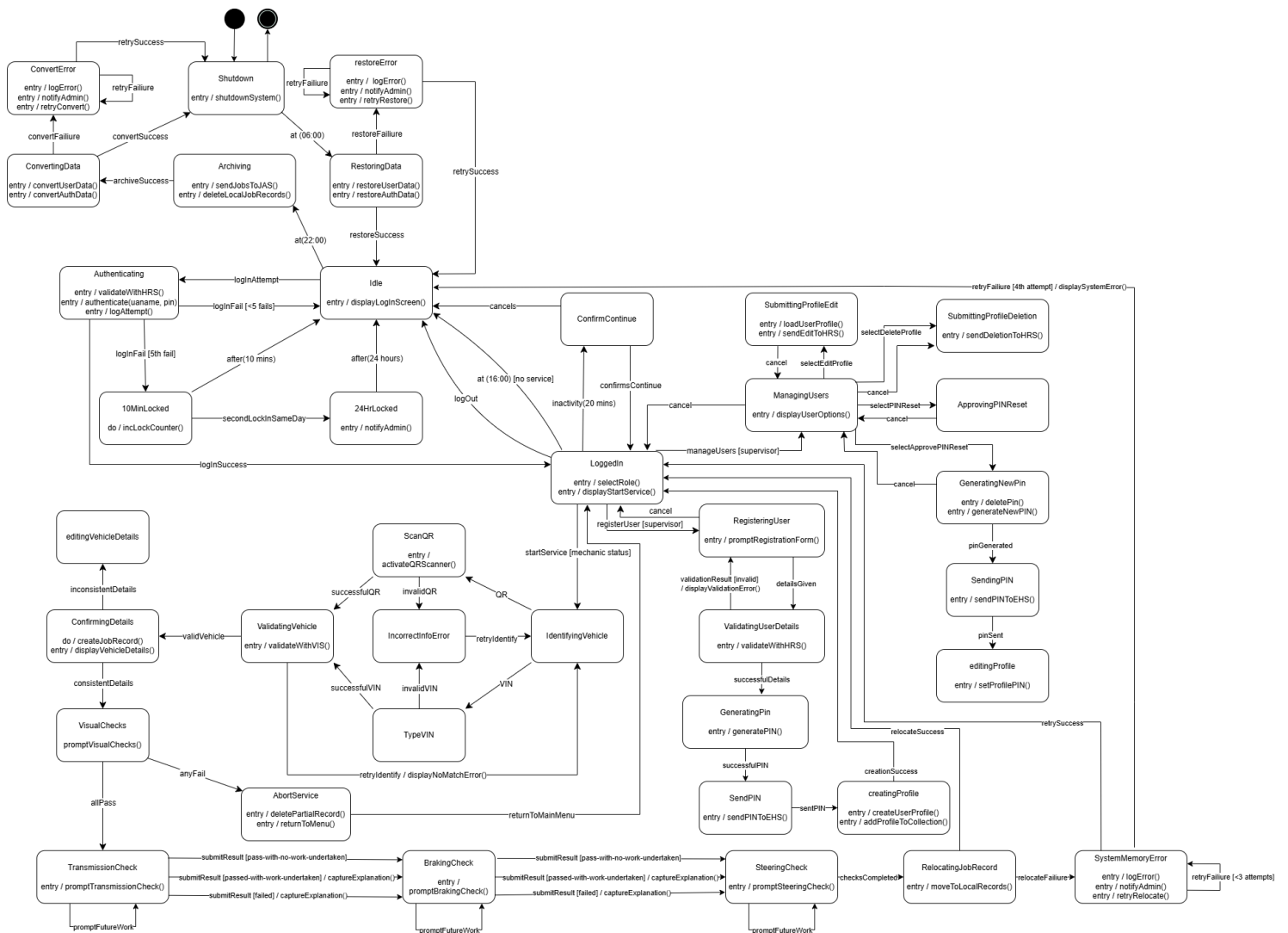
Top Half



Bottom Half

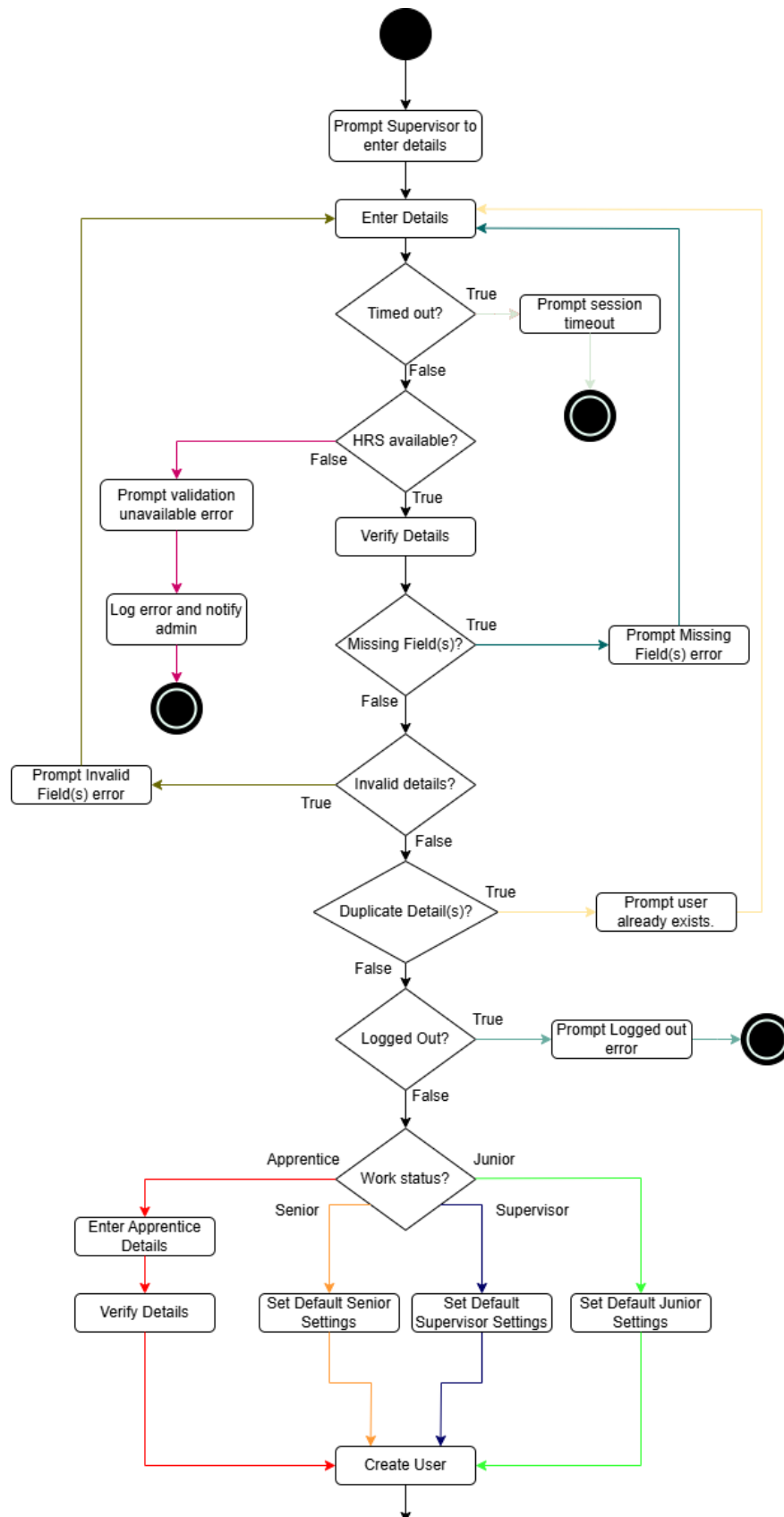


D9 – State Machine Diagram

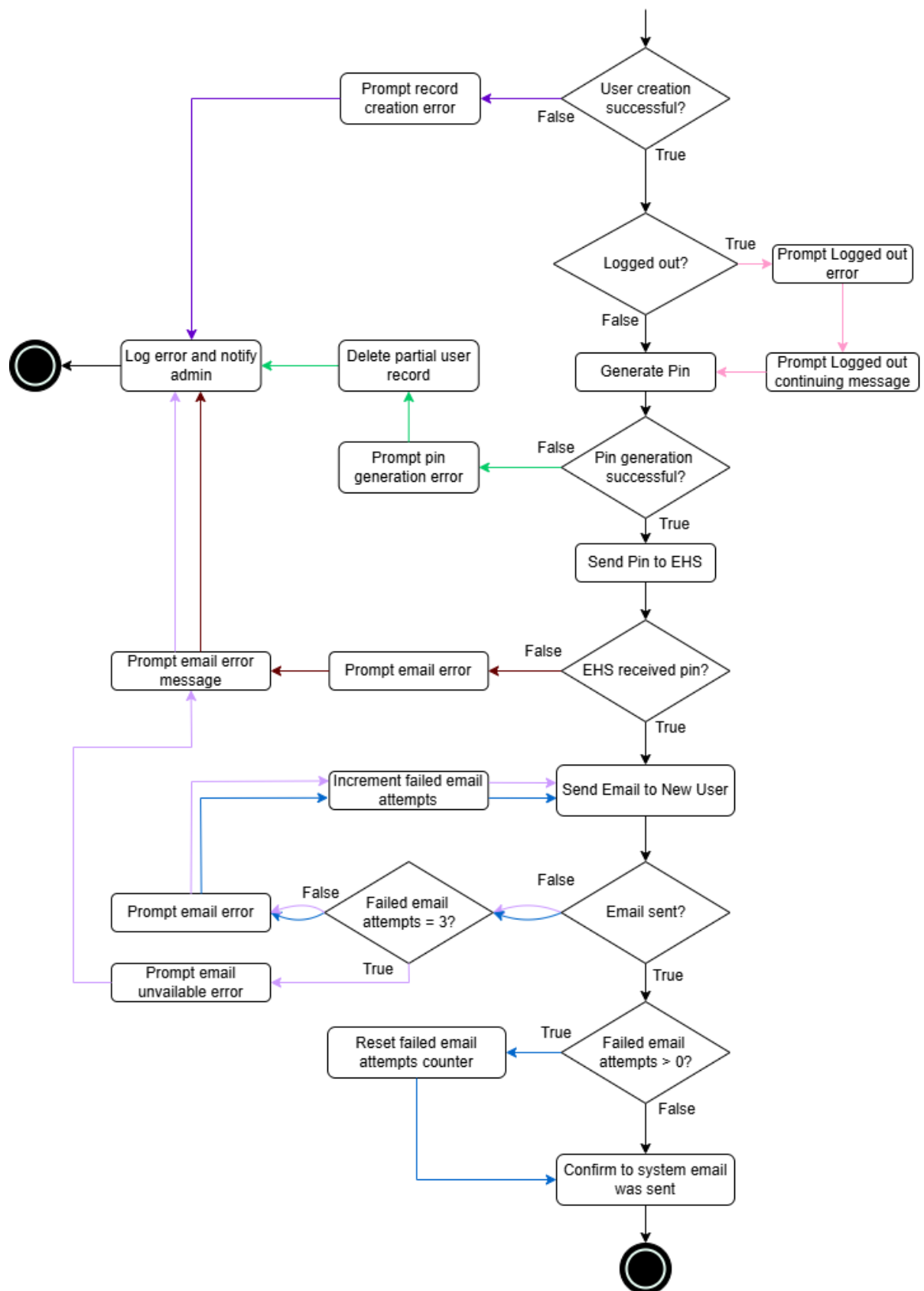


D10 – Flow Graph and Scenario Tests

D10.1 - Top half



D10.1 - Bottom Half



To avoid creating 18 separate flow graphs for each path, all paths are represented within a single diagram using different colours to represent specific edges. Each path initially follows the main flow (black) until it diverges at its corresponding coloured section, which represents its unique scenario. **Note: For scenarios where work status is not relevant (paths 5-18), any work status path may be taken.**

General Assumptions:

- Supervisor is logged in.
- Supervisor is on user registration page.
- Session is active.
- All user details entered are unique and valid.
- HRS is available.
- User creation is successful.
- PIN generation is successful.
- EHS successfully receives PIN.
- Email is successfully sent.
- Failed email attempts is zero.

D10.2 - Scenario Tests

Path	Scenario Preconditions	Scenario Inputs	Expected Outcome
1	• Work status selected is Apprentice	Supervisor enters user registration details Supervisor enters Apprentice Details.	A new user of status Apprentice is made and the new user receives a confirmation email along with their assigned PIN.
2	• Work status selected is Junior.	Supervisor enters user registration details	A new user of status Junior is made and the new user receives a confirmation email along with their assigned PIN.
3	• Work status selected is Senior.	Supervisor enters user registration details.	A new user of status Senior is made and the new user receives a confirmation email along with their assigned PIN.
4	• Work status selected is Supervisor.	Supervisor enters user registration details.	A new user of status Supervisor is made and the new user receives a confirmation email along with their assigned PIN.
5	• Session times out during registration process.	Supervisor enters user registration details.	Supervisor is timed out of session and registering user is terminated.
6	• HRS is unavailable.	Supervisor enters user registration details.	As HRS is unavailable, the supervisor is displayed with an error, the error is logged, admin is notified and registering user is terminated.

7	Supervisor enters missing field(s).	Supervisor enters user registration details with missing field(s). Supervisor enters user registration details.	After the Supervisor initially entered details with missing field(s), they are notified of this error and they retry successfully. A new user is made and the new user receives a confirmation email along with their assigned PIN.
8	Supervisor enters invalid detail(s).	Supervisor enters invalid user registration details. Supervisor enters user registration details.	After the Supervisor initially entered invalid detail(s), they are notified of this error and they retry successfully. A new user is made and the new user receives a confirmation email along with their assigned PIN.
9	Supervisor enters duplicate detail(s).	Supervisor enters user registration details with duplicate username. Supervisor enters user registration details.	After the Supervisor initially entered the username of a user that already exists, they are notified of this error and they retry successfully. A new user is made and the new user receives a confirmation email along with their assigned PIN.
10	Supervisor is logged out before new user is created.	Supervisor enters user registration details.	As Supervisor is logged out before the new user is created, they are prompted with a log out error and registering user is terminated.
11	User creation is unsuccessful.	Supervisor enters user registration details.	As user creation is not successful, Supervisor is prompted with record creation error, the error is logged, the admin is notified and registering user is terminated.
12	Supervisor is logged out after user is created.	Supervisor enters user registration details.	As Supervisor is logged out after the new user is created, the user is prompted with a log out error and they are told not to re-register the new user and check email for confirmation and PIN. They are also told if they have not received an email, to contact IT services. The process of registering a user is attempted to be completed in the background.
13	PIN generation fails	Supervisor enters user registration details.	As PIN generation fails, Supervisor is prompted with a PIN generation error, the partial user record is deleted, error is logged, admin is notified and registering user is terminated.
14	EHS fails to receive generated PIN.	Supervisor enters user registration details.	As EHS unsuccessfully received newly generated PIN, Supervisor is prompted with an email error and they are told that the new user has been successfully created and to wait for future emails with PIN and confirmation. The error is then logged, admin is notified and registering user is terminated.

15	Email fails to send once.	Supervisor enters user registration details. Supervisor signals email failed to send.	As the email failed to send once, Supervisor is notified with an email error, failed email attempts is incremented and a successful attempt to resend the email was made. The failed email attempts counter was reset before finishing user registration. A new user of status Apprentice is made, and the new user receives a confirmation email along with their assigned PIN.
16	Email fails to send twice.	Supervisor enters user registration details. Supervisor signals email failed to send. Supervisor signals email failed to send.	As the email failed to send twice, Supervisor is notified with an email error each time, failed email attempts is incremented each time and then a successful attempt to resend the email was made. The failed email attempts counter was reset before finishing user registration. A new user of status Apprentice is made, and the new user receives a confirmation email along with their assigned PIN.
17	Email fails to send three times.	Supervisor enters user registration details. Supervisor signals email failed to send. Supervisor signals email failed to send. Supervisor signals email failed to send.	As the email failed to send three times, Supervisor is notified with an email error each time, failed email attempts is incremented each time and then a successful attempt to resend the email was made. The failed email attempts counter was reset before finishing user registration. A new user of status Apprentice is made, and the new user receives a confirmation email along with their assigned PIN.
18	Email fails to send four times.	Supervisor enters user registration details. Supervisor signals email failed to send. Supervisor signals email failed to send. Supervisor signals email failed to send. Supervisor signals email failed to send.	As the email failed to send four times, Supervisor is notified with an email error the first three times, failed email attempts is incremented the first three times. On the fourth attempt to resend the email, Supervisor is prompted with an email unavailable error, and they are told that the new user has been successfully created and to wait for future emails with the confirmation and PIN email.